# UNSUPERVISED MACHINE LEARNING FOR SATELLITE IMAGE CLASSIFICATION

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## 1 Introduction

Unsupervised machine learning (ML) is a framework which learns patterns from unlabelled data and has a potential use as a method to classify different landscape features of a satellite image. A satellite image of Nicaragua's Momotombo volcanic eruption in 2015 represents the information captured from Earth's surface via sensors on the satellite. The images are composed of spectral bands, which give measurements of reflectance at differing wavelengths and therefore enable identification of terrain features. This concept will be used to attempt to classify features of the volcanic eruption satellite image into 4 key features; forests, clouds and smoke, old/cold lava flows and new/hot lava flows.

The spectral bands provided to investigate these features are; blue, green, red, near infrared ('nir'), short wavelength infrared ('swir1', 'swir2') and thermal.

## 2 Methodology

### 2.1 Preprocessing

The satellite image is cropped to isolate the key features. This focusses the ML algorithm on the chosen region containing meaningful variations, rather than including noisy or vague areas which affect the accuracy of the clustering. The chosen region is shown in Figure 1. Histograms for each of the 7 spectral bands are plotted for an initial look at the distribution of the data. The bands all exhibit similar numerical distributions apart from the 'thermal' band, with a mean value around $10^3$ larger than the other bands. To account for this difference, scikit-learn[2] 'StandardScaler' is used to standardise all features by removing the mean and scaling to unit variance.

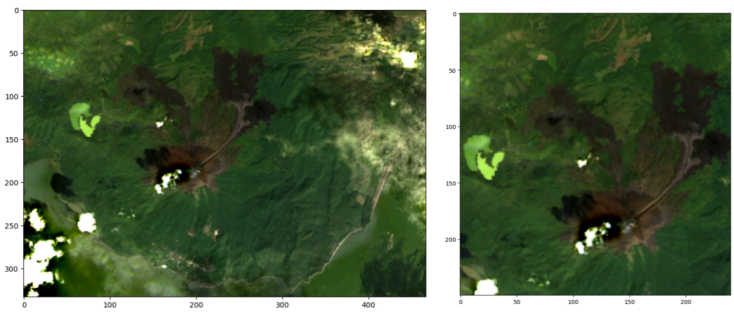

Figure 1: *Original satellite image (left) and preprocessed cropped image (right).*

### 2.2 Principle Component Analysis (PCA) and Independent Component Analysis (ICA)

To improve the quality of the input data to the model, PCA or ICA are techniques which can be used. PCA is a dimensionality reduction method which retains a specified amount of principle components and reduces the dimensionality of the dataset. This reduces noise and makes the subsequent clustering more robust. A smaller dataset is easier to explore and visualise. In summary, the aim of PCA is to reduce the number of variables in a dataset whilst retaining as much information as possible.

Independent Component Analysis (ICA) is a statistical technique used in analysing non-Gaussian signals and separating them into independent sources[1]. The main purpose is to identify the underlying structure of a signal and extract the constituent components. These components are typically assumed to be non-correlated and non-Gaussian.

### 2.3 K-means clustering

K-means clustering is suitable for this task due to the ability of the method to group similar spectral features or regions (e.g., clouds, land types) without any prior knowledge of their labels. A user-specified number of clusters split the data into K groups of equal variance. Using the Euclidean distance the model finds the closest points to the centroids. This method assumes similarly sized, isotropic clusters with equal variance.

### 2.4 Model Evaluation

Evaluating the ML model is necessary to ensure that the model is reliable, generalisable, and capable of making accurate predictions on unseen data. Overfitting and underfitting (e.g. model is too complex or too simple, respectively) are consequences of an ML model with poor performance. There are many different metrics that can be used to evaluate a model. The primary evaluation of this model will be via direct comparison of the clustered image and the original image in Figure 1. For a quantitative metric alongside qualitative interpretation, scikit learn[2] 'silhouette score' will be used.

The silhouette score measures how well-defined clusters are by evaluating the cohesion within clusters and the separation between clusters. It is an appropriate metric for this unsupervised task, as it provides a quantitative assessment of the quality of clustering without requiring labelled data. The score ranges from -1 to 1, with -1 meaning data is likely assigned to the wrong cluster and 1 meaning data is well clustered and far from neighbouring clusters.

## 3 Results and Discussion

K-means clustering is run 3 times; on raw data, on data preprocessed with PCA and on data preprocessed with ICA. A silhouette score is calculated for each clustering attempt. 5 clusters minimum are chosen as a result of plotting cumulative explained variance ratio, showing that 5 clusters explains more than 99% of the data.

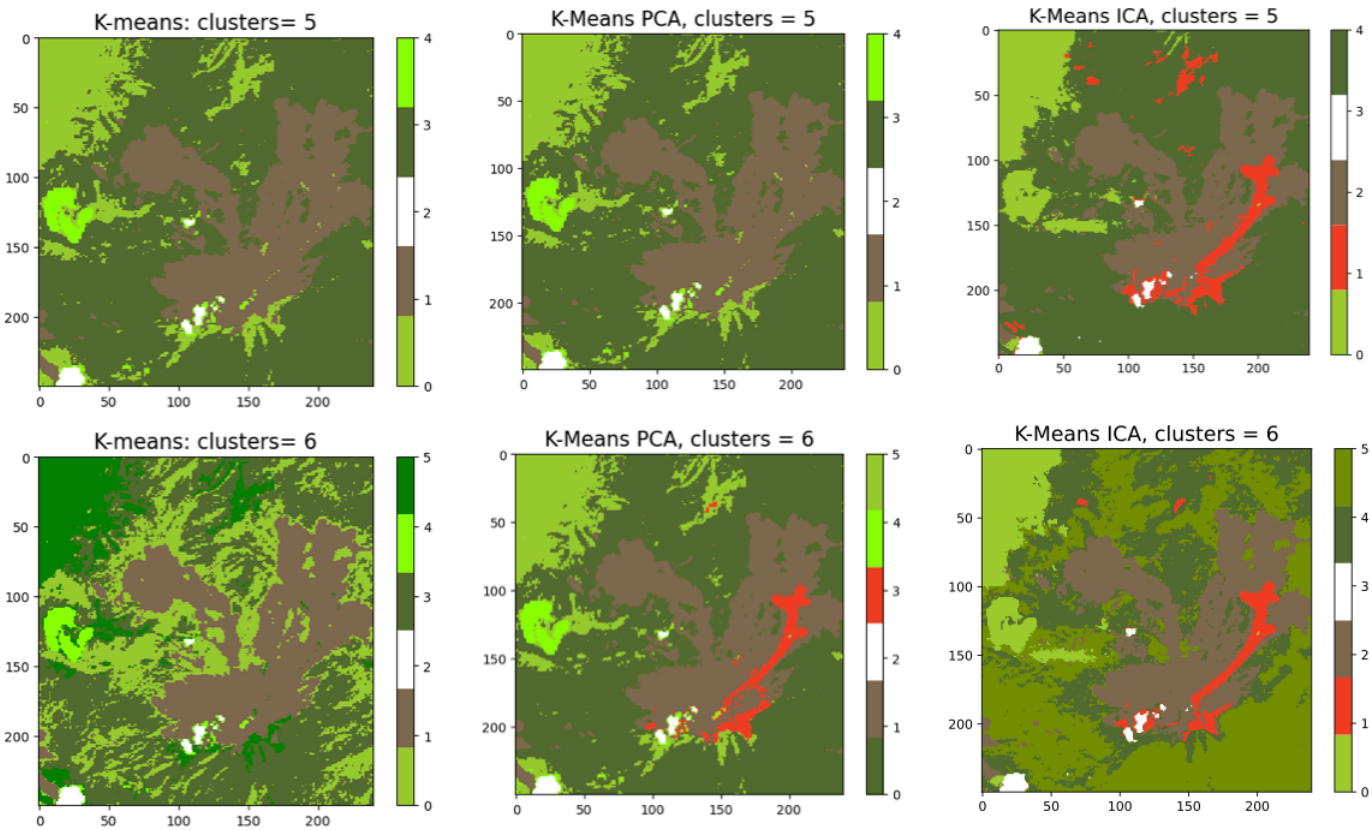


Figure 2: *K-means algorithm applied to: raw data (left), data with PCA (middle), and data with ICA (right) for 5 clusters (top) and 6 clusters (bottom). Clusters are coloured according to feature represented: green tones=forest or lake, brown=old lava flow, red=new lava flow and white=cloud/smoke.*

| Clustering Algorithm Silhouette Scores | | |
|---|---|---|
| Model | 5 Clusters | 6 Clusters |
| Raw Data | 0.436 | 0.258 |
| With PCA | 0.441 | 0.447 |
| With ICA | 0.402 | 0.272 |

Table 1: *Silhouette scores for clustering models with 5 and 6 clusters, comparing results for raw data, PCA, and ICA preprocessing.*

K-means on raw data in Figure 2 shows a separation into clouds, old lava flow and forest, also identifying the algae covered lake as a separate cluster. 6 clusters finds more detail in the forest area around the eruption. There is no indication, however, of the thermal band corresponding to the hot lava of the volcanic eruption. To try and account for this important feature, the preprocessing stage of the pipeline can be modified. The thermal band weight is doubled to see if the K-means algorithm can pick out this feature, but no different results are seen for the raw data.

PCA is carried out to aid the algorithm by reducing redundancy via transforming the data into a set of uncorrelated components, where each PC captures a unique, orthogonal aspect of the image's spectral content. 5 clusters separate all features apart from the new lava flow. 6 clusters finds all features, including the new lava flow and therfore successfully identifies all required features.

The results for ICA preprocessing cluster the features successfully for both 5 and 6 clusters. The algae-covered lake is not identified as separate cluster as per all other results, but this is not a key feature of the image and suggests that this method is primarily finding more important features. 6 clusters finds a good description of the new flow identified in the thermal band, however mistakes some of the more reflective surrounding area as new flow which PCA does not do. More detail in the forest area is shown for 6 clusters, potentially due to the differences which arise from dead vegetation from previously burnt land area. Further investigation would decide if the detail of 6 clusters in comparison to 5 is required.

PCA or ICA are required to capture all key features of the landscape in this image. Preprocessing with ICA requires less clusters to capture every key feature. A silhouette score calculated for each model provides insight into the clustering ability. Table 1 shows that values for all models are greater than 0, indicating acceptable clustering. 6 clusters with PCA preprocessing has a higher silhouette score than 5 clusters, confirming the previous qualitative analysis. Silhouette scores for ICA models would indicate somewhat poorer clustering ability for 6 clusters vs 5 with this method, however the key features are most successfully identified with this model.

The clusters can provide more insight into the nature of the physical properties of the satellite image and how this influences the K-means clustering. To investigate this, a plot of spectral band type vs mean reflectance can be plotted.

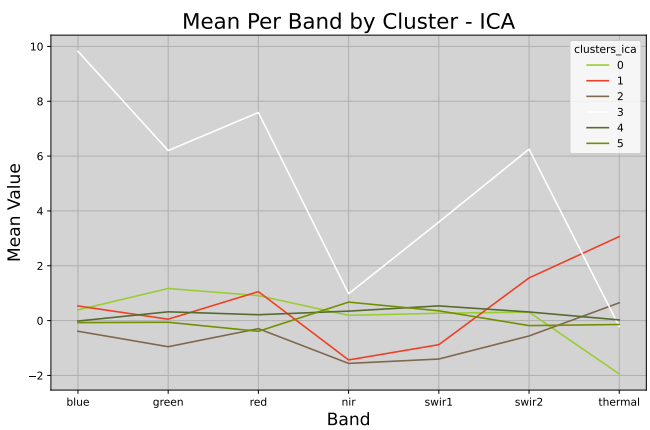


Figure 3: *Mean per band per cluster for ICA model with 6 clusters. Colours represent features of satellite image (white=clouds, green tones = forests, red=new lava, brown=old lava.)*

The green trends do not vary greatly across all bands, which implies these features do not vary considerably within the cluster. Wide gaps between cluster lines suggest better-defined clusters and more distinct feature importance. More variability is seen across the white, red and brown trends representing the clouds, new lava flow and old lava flow, suggesting that these bands strongly contribute to specific clusters. The white and red trends have a similar pattern across all bands except the thermal which could suggest this is an important band for discriminating between clouds and new lava flow. The white trend shows a high level of reflectance in the visible bands due to scattering, but low levels of reflectance in 'nir' and thermal. Figure 3 could also explain clustering discrepancies in the cloud-covered region, especially for the surface directly beneath. The shadows cast by clouds cause an issue for clustering as these are incorrectly identified as old lava flows due to their similar dark properties. Clouds may also not completely block the satellite sensor's view of the underlying terrain and influence the spectral readings. Environmental effects must be accounted for in the development of such models to ensure accurate results.

# 4 Conclusion

K-means clustering has shown to be an effective tool for the task of separating a Landsat8 image into different geographical features. For K-means PCA with 6 clusters, and K-means ICA for both 5 and 6 clusters, all of the key features specified in Section 1 can be identified separately. This approach has limitations as discussed in Section 3; features of the satellite image can influence inaccurate clustering. The shadows resulting from clouds are misidentified as old lava flows, implying a greater burnt area. The angle which the satellite image is taken at and time of day would also affect the shadow effects. The implications of this should be considered in further work. The thermal band indicating new lava flows was not seen as a separate cluster when K-means was carried out on raw data. This is an important feature and so PCA or ICA prior to K-means clustering are essential preprocessing measures. Although the PCA method can successfully split the features, this assumes Gaussian data which is rarely the case in these images. ICA is preferable due to the assumption of non-Gaussian data.

A silhouette score provided further insight into the model's clustering ability. In this scenario however, qualitative analysis from direct comparison with Figure 1 and the spectral band images are more valuable. This shows that meaningful human interpretation of, and input to, ML models is not to be overlooked despite the importance of having performance evaluation metrics. In further work and for confirmation of these findings, a supervised method could be employed to identify a label for each pixel to compare clusters for a more robust conclusion of the accuracy of the model. This may require a survey carried out from visiting the area and confirming the status of different parts of the land.

Refinement of this method is shown to be required. As previously mentioned, a supervised learning method could be used to train the K-means algorithm. Euclidean distance is used in the K-means algorithm in this work, but work in a similar task has investigated using different distance metrics such as Manhattan distance and Chebyshev distance[3] which could be investigated in future work as an area of improvement. Efficiency of the model could also be improved. It may be the case that some of the spectral bands are not required to successfully identify key features, and these could be discarded for computational efficiency. An investigation into the weighting of more important bands, as was briefly touched upon in this work, would also provide further insight.

Real-time segmentation of natural events such as fire, floods and volcanoes is essential. If the severity and location of these events could be determined by a satellite as soon as they occur, lives and resources could be saved. This work shows promise for the use of ML methods for real-time segmentation, highlighting areas of improvement for future work.

# References


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[3] Aleksei Rozanov. Semantic segmentation of remote sensing imagery using k-means, Mar 2024.